higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

DIGITAL ELECTRONICS N4

15 August 2025

This marking guideline consists of 6 pages.

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NATED Question Papers	**N1-N6** Mathematics NATED	**N1-N4** Engineering Science NATED	**N1-N6** Industrial Electronics NATED	**N4-N6** Electrotechnics NATED	**N4-N6** Mechanotechnics NATED
N5-N6 Power Machines NATED	**N4-N6** Quantity Surveying NATED	**N5-N6** Strength of Materials & Structures NATED	**N1-N3** Engineering Drawing NATED	**N3** Eletro-technology NATED	**N3** Mechano-technology NATED
N4-N6 Building Administration NATED	**N4-N6** Building& Structural Surveying NATED	**N4-N6** Building & Structural Construction NATED	**N1-N3** Building Science NATED	**N1-N3** Building Drawing NATED	**N4** Mechanical Draughting NATED
N1-N3 Electrical Trade Theory NATED	**N2-N3** Diesel Trade Theory NATED	**N5-N6** Strength of Materials & Structures NATED	**N3** Building & Civil Technology NATED	**N1-N2** Fitting and Machining Theory NATED	**N1-N4** Plating and Structural Steel Drawing NATED
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QUESTION 1

1.1
1.1.1 C✓
1.1.2 B✓
1.1.3 D✓
1.1.4 C✓
1.1.5 B✓

(5 × 1) (5)

1.2
1.2.1 False✓
1.2.2 True✓
1.2.3 True✓
1.2.4 True✓
1.2.5 True✓

(5 × 1) (5)
[10]

QUESTION 2

2.1 2.1.1 14, 5_8 = $1100,101_2$ ✓ & 7, B_{16} = $0111,1011_2$ ✓

one's complement of 0111,1011 = 1000,0100✓

```
  0 * 1 1 0 0 1 0 1 0
      ,
  1 * 1 0 0 0 0 1 0 0
      ,
1 0 * 0 1 0 0 1 1 1 0   ✓✓
      ,
```

(6)

Ignore the carry and the answer is positive. $100,111_2$ = $4,875_{10}$✓

2.1.2 2C, 5_{16} = $101100,0101_2$ ✓ & 6, 25_{10} = $110,01_2$ ✓

```
                      1  1  1,  0  0  ✓
  1  1  0  0  1 | 1  0  1  1  0  0  0  1,  0  1
                  1  1  0  0  1  ↓  ↓  ↓  ↓  ✓
               ---------------------------------
                  1  0  0  1  1  0  ↓  ↓  ↓
                     1  1  0  0  1  ↓  ↓  ↓
                  ------------------------------
                     0  1  1  0  1  1  ↓  ↓  ✓
                        1  1  0  0  1  ↓  ↓
                     ---------------------------
                        0  0  0  1  0  0  ↓
                                 0  0  0  1
                     ---------------------------
```

$111,000_2$ = 7.0_8 ✓ (6)

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2.1.3 12, 125_{10} = 1100, 001_2✓ & 5, 4_8 = 101, 1_2✓

			1	1	0	0	0	0	1		
			X			1	0	1	1		
			1	1	0	0	0	0	1		✓
		1	1	0	0	0	0	1			
	1	1	0	0	0	0	1				✓
1	0	0	0	0	1	0	1	0	1	1	✓

$1000010,1011_2 = 42, B_{16}$✓ (6)

2.2 2.2.1 $11110010,101_{grey}$ to Octal number system.

1	1	1	1	0	0	1	0,	1	0	1
↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓
1	0	0	0	1	0	1	1,	1	1	1 ✓

$10001011,111_2 = 010\ 001\ 011,\ 111 = 213,7_8$✓✓ (3)

2.2.2 11001011100xs-3 code

0110	0101	1100	
↓	↓	↓	
6	5	12	✓
-3	-3	-3	
3	2	9	✓

decimal equivalent of 11001011100xs-3 code = 329_{10}✓ (3)
[24]

QUESTION 3

3.1 3.1.1 A✓
3.1.2 XY✓
3.1.3 A✓

(3 × 1) (3)

3.2 3.2.1 $A(A+B)(\overline{A}+\overline{B}+C)=F$

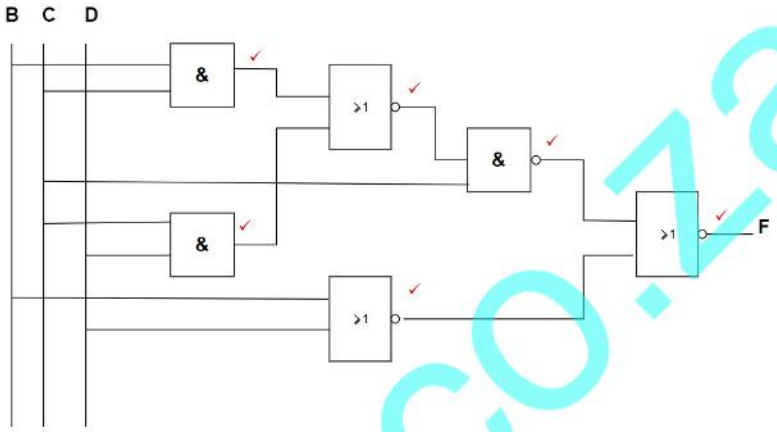
$A+AB(\overline{A}+\overline{B}+C)$ ✓✓
$A(1+B)(\overline{A}+\overline{B}+C)$
$A(\overline{A}+\overline{B}+C)$ ✓
$A\overline{B}+AC$
$A(\overline{B}+C)$ (6)

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3.2.2 $Z\left(X+\overline{Y}\right)+X\left(Y+\overline{Z}\right)+\overline{X}\,Y=\ F$

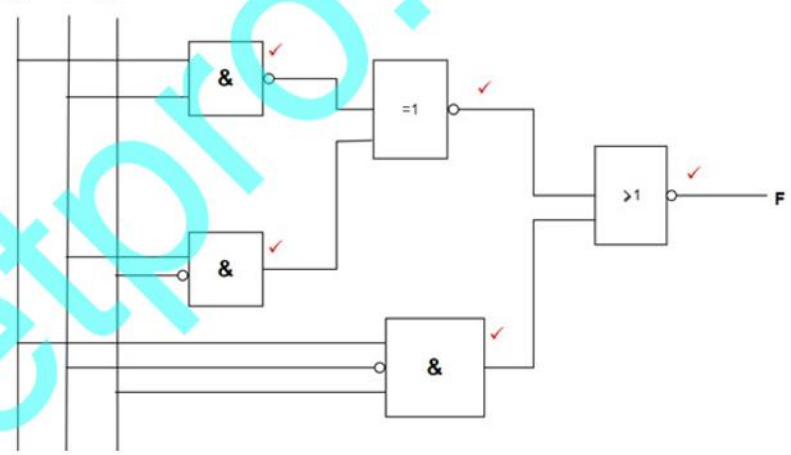
$XZ+\overline{Y}\,Z+XY+X\,\overline{Z}+\overline{X}\,Y$
$X\left(Z+\overline{Z}\right)+Y\left(X+\overline{X}\right)+\overline{Y}\,Z$
$X+Y+\overline{Y}\,Z$
$X+Y+Z$ (6)

3.3 3.3.1 $\overline{\overline{\overline{\left(BC+CD\right)}\ C}+\overline{B+D}}=F$



(6)

3.3.2 $\overline{\overline{X\overline{Y}\ \oplus\ Y\,\overline{Z}}+X\,\overline{Y}\,Z}=F$

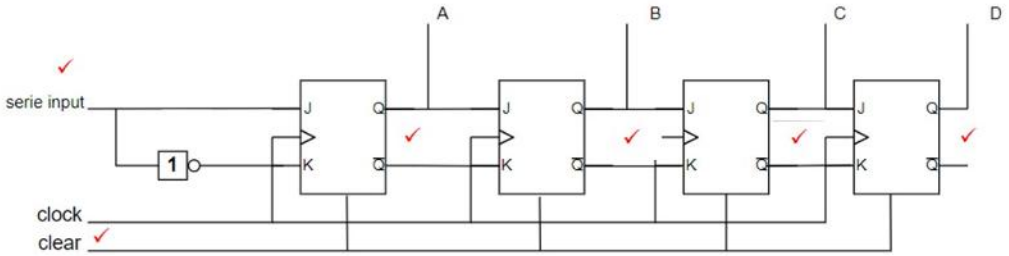


(5)
[26]

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QUESTION 4

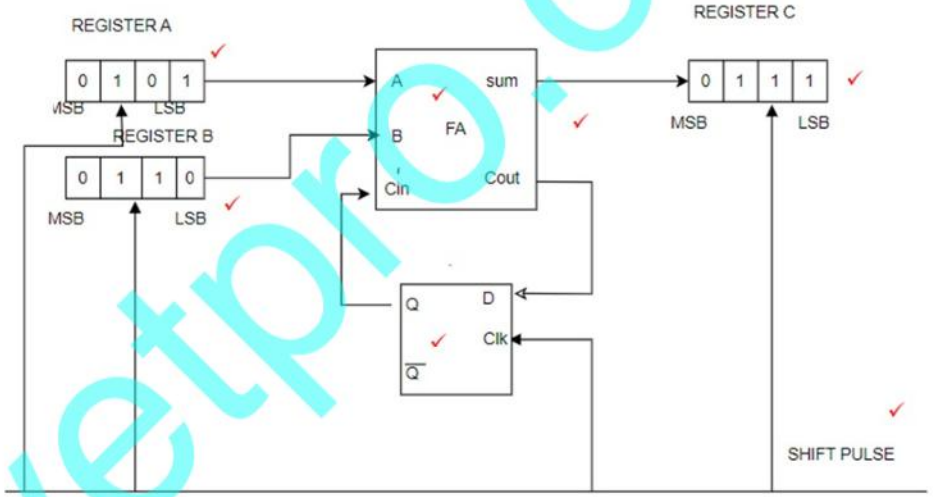
4.1



JA	A	B	C	D
0	0	0	0	0
1	1	0	0	0
0	0	1	0	0
0	0	0	1	0
0	0	0	0	1
0	0	0	0	0

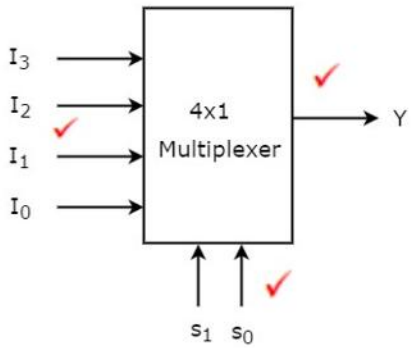
(10)

4.2



(7)

4.3

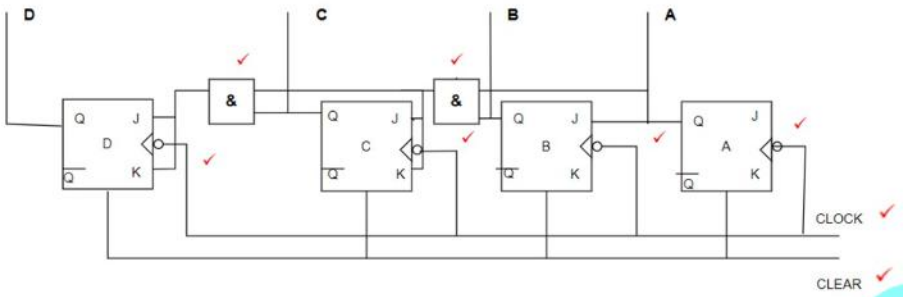


(3)
[20]

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QUESTION 5

5.1



(8)

5.2 TWO advantages of COMS

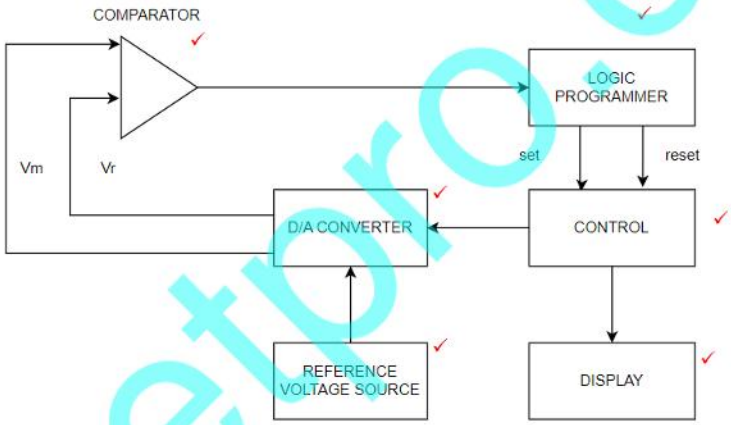
- Improved switching speed✓
- Lower power dissipation✓

(2)

5.3 5.3.1 Noise immunity is the amount of noise on the input of a logic gate without affecting the output of logic state (2)

5.3.2 Logic swing is the voltage difference between the logic levels for 1 and 0. (2)

5.4



(6)
[20]

TOTAL: 100

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